16. Small tomato borer (*Neoleucinodes elegantalis*)

Last modified: October 16, 2026

16A. Introduction

The small tomato borer (STB), *Neoleucinodes elegantalis* (Guenée) (Lepidoptera: Pyralidae), is an oligophagous pest of Solanum species, including tomato, eggplant, red and green pepper, lulo (naranjilla), and wild Solanum spp. (Molet 2012, OEPP/EPPO 2015). Native to South America, it has expanded its range into Mexico, Central America, and the Caribbean (Molet 2012, OEPP/EPPO 2015, Díaz-Montilla 2013a). The species occupies a wide range of climatic conditions in South America, although its distribution varies depending on host plant availability and altitude (Díaz Montilla 2013b). Larvae develop through five instars, with the third through fifth instars responsible for the majority of fruit damage, and development rates that may vary by host plant (Serrano Plaza et al. 1992, Díaz-Montilla 2013a). Infestations can result in severe economic losses, with reported yield reductions in South American tomato crops ranging from 45% to 90% (Díaz Montilla and Kondo 2024). If introduced to the United States, regions with high levels of tomato and pepper production would be particularly vulnerable (Molet 2012, USDA-APHIS-PPQ 2024). In addition, STB poses a significant barrier to the export of solanaceous crops from South America to the United States and the European Union (Díaz-Montilla 2013a, Díaz-Montilla 2013b). The pest has been intercepted hundreds of times at United States ports of entry, with most interceptions originating from Brazil (Molet 2012).

Potentially suitable environmental conditions for STB occur in western portions of the Pacific states, the Southeast, and parts of the southern Northeast, although colder conditions limit suitability across much of the northern United States (Barker et al. 2020). In its native range, increases in temperature and precipitation appear to favor STB populations, while arid conditions can limit its distribution, suggesting that moisture availability may influence STB distribution and climatic suitability in the contiguous United States (CONUS; Silva et al. 2020, da Silva et al. 2018). Annual voltinism also varies substantially with climate, with models predicting 2–11 generations per year in Colombia and 5–10 generations in Brazil (de Moraes and Foerster 2015, dos Santos and Marchioro 2021). STB does not appear to have a diapause or distinct overwintering stage and can produce continuous, overlapping generations, with voltinism generally increasing toward warmer, lower-latitude regions (Serrano Plaza et al. 1992, Moraes and Foerster 2015, Barker et al. 2020). In CONUS, annual voltinism is predicted to range from one to two generations in colder parts of the Pacific Northwest to seven to nine generations in coastal areas of the Southeast, with three to six generations across much of the remainder of the CONUS (Barker et al. 2020).

A model for tomato fruit borer developed for the Degree-Day, establishment Risk, and Phenological event mapping system (DDRP) is operationalized to produce near real-time forecasts for the CONUS at https://uspest.org/DDRP. The phenology model applies a lower developmental threshold of 8.89 °C and upper threshold of 30 °C for all life stages (egg, larva, pupa, and adult) and assumes the adult is the overwintering stage. The climatic suitability model applies a lower cold stress threshold and upper heat stress threshold of 6 °C and 32 °C, respectively. The full set of model parameters for tomato fruit borer, including degree-day requirements for individual life stages and climate stress limits, are presented in Table 1. Details on model parameterization are available at USPest.org (Barker et al. 2020).

Table 1. STB model parameters for DDRP.

Parameter	Code	Value
Lower developmental thresholds (°C)		

Parameter	Code	Value
Egg	eggLDT	8.89
Larvae	larvaeLDT	8.89
Pupae	pupaeLDT	8.89
Adult	adultLDT	8.89
Upper developmental thresholds (°C)		
Egg	eggUDT	30
Larvae	larvaeUDT	30
Pupae	pupaeUDT	30
Adult	adultUDT	30
Stage durations (°C degree-days)		
Egg	eggDD	86
Larvae	larvaeDD	283
Pupae	pupDD	203
Adult	adultDD	96
Pest events (°C degree-days)		
J-larval development complete	OWEventDD	86
Egg hatch	eggEventDD	80
Developing J-larvae	larvaeEventDD	140
Adult emergence (pupal development complete)	pupaeEventDD	100
Egg laying	adultEventDD	55
Cold stress		
Cold stress temperature threshold (°C)	coldstress_threshold	2
Cold degree-day (°C) limit when most individuals die	coldstress_units_max1	550
Cold degree-day (°C) limit when all individuals die	coldstress_units_max2	900
Heat stress		
Heat stress temperature threshold (°C)	heatstress_threshold	32
Heat stress degree-day (°C) limit when most individuals die	heatstress_units_max1	180
Heat stress degree-day (°C) limit when all individuals die	heatstress_units_max2	340
Cohorts		
Degree-days (°C) to emergence (average)	distro_mean	50
Degree-days (°C) to emergence (variation)	distro_var	1500
Minimum degree-days (°C) to emergence	xdist1	0
Maximum degree-days (°C) to emergence	xdist2	111
Shape of the distribution	distro_shape	normal
Other		
Order of stages	stgorder	OA, E, L, P, A
Degree-day calculation method	calctype	triangle

16B. Results

i. Phenology

Entire period: 1981–2025

Over 1981–2025, predicted adult emergence and egg hatch generally occurred earlier across much of the southern and eastern United States, while later events were evident across portions of the northern and western interior (Figure 1, Figure 2). Significant changes were concentrated in the eastern and central United States.

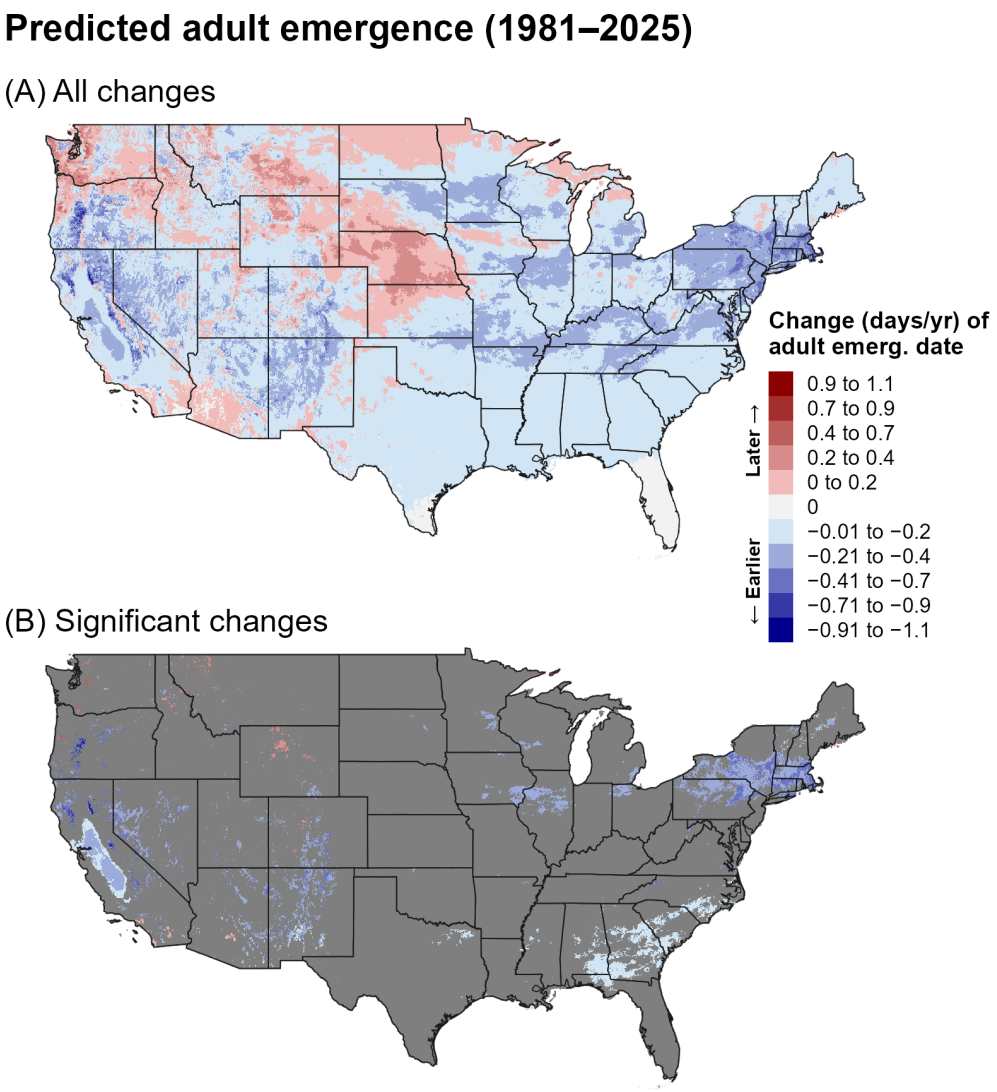


Figure 1. Changes (days/year) in predicted dates of first adult emergence for STB across the CONUS over the entire 44-year period (1981–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant.

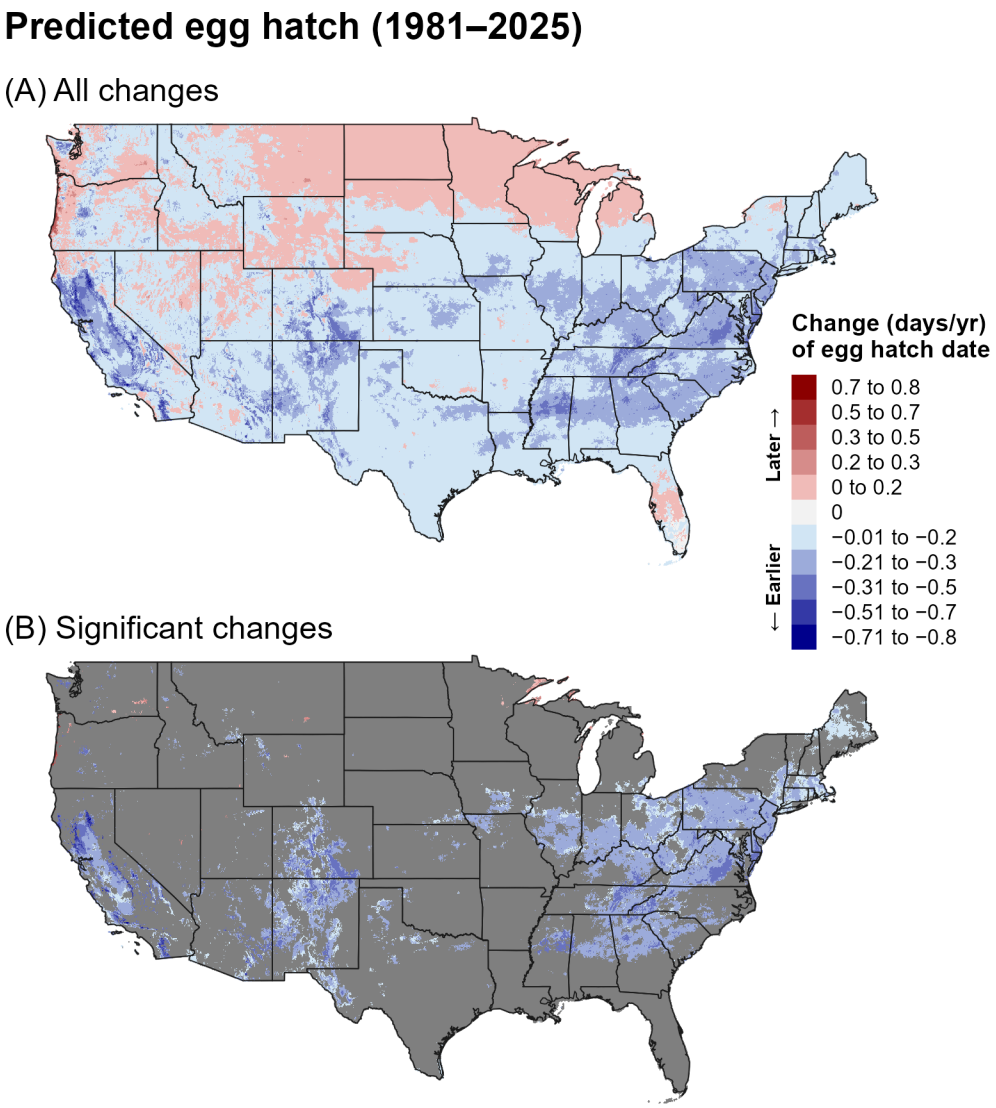


Figure 2. Changes (days/year) in predicted dates of first egg hatch for STB across the CONUS over the entire 44-year period (1981–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant.

Recent period: 2001–2025

During 2001–2025, adult emergence and egg hatch showed a stronger tendency toward later dates across portions of the northern and central United States, with earlier events remaining in scattered western and southern areas (Figure 3, Figure 4). Significant changes were relatively localized, with some concentrated areas in the central United States.

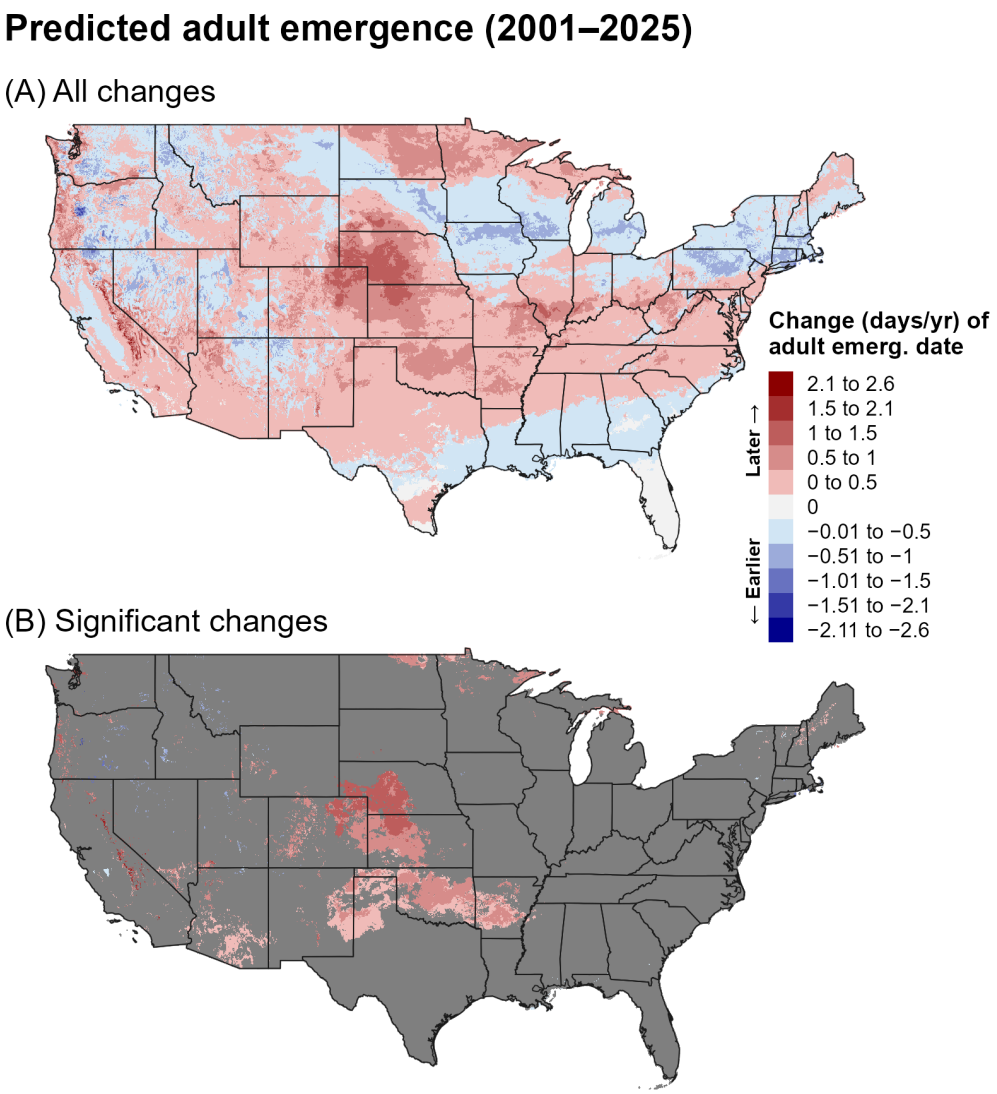


Figure 3. Changes (days/year) in predicted dates of first adult emergence for STB across the CONUS over the recent 24-year period (2001–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant.

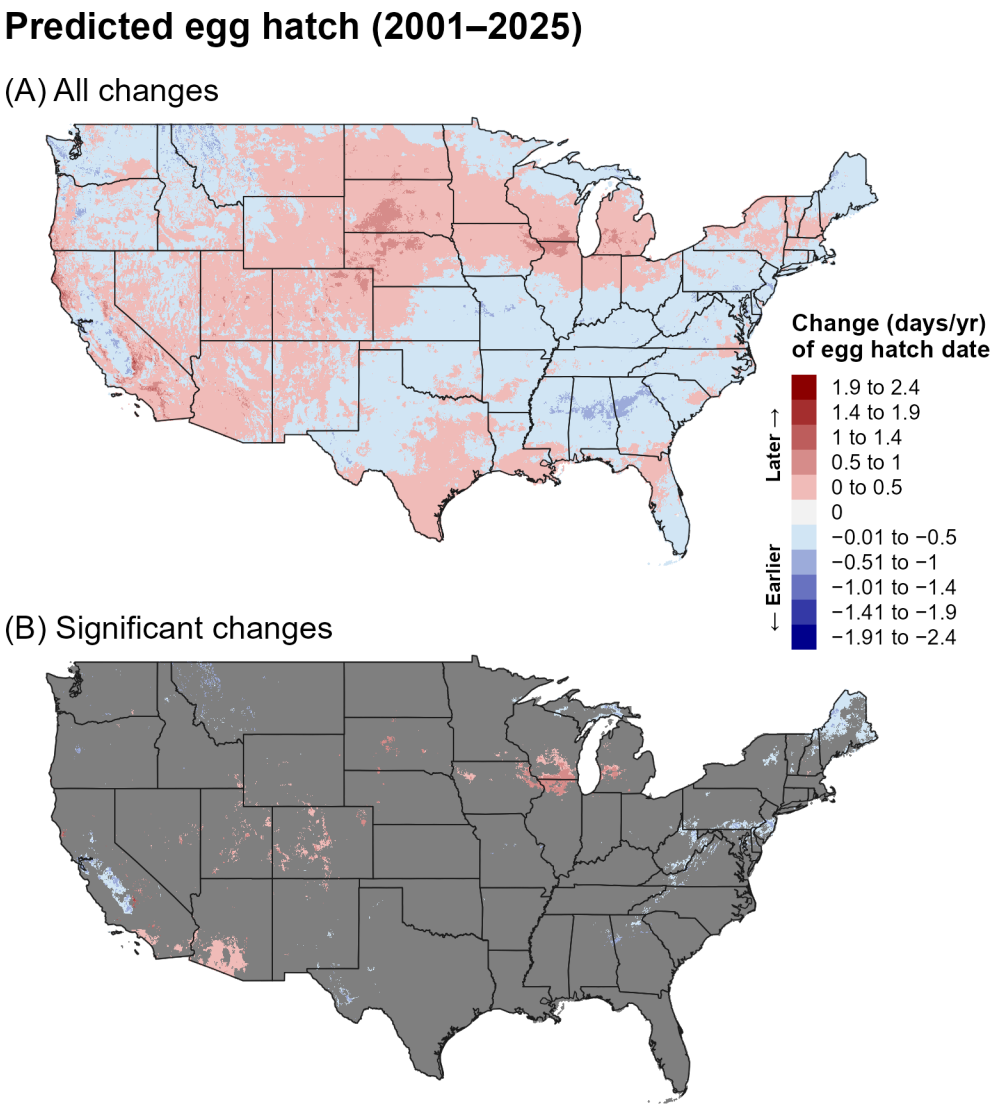


Figure 4. Changes (days/year) in predicted dates of first egg hatch for STB across the CONUS over the recent 24-year period (2001–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant.

ii. Climatic suitability

Entire period: 1981–2025

Predicted cold stress decreased across broad portions of the northern United States over 1981–2025, while heat stress increased across much of the western and southern United States (Figure 5, Figure 6). Predicted climatic suitability increased significantly in portions of the Great Lakes, Northeast, and central United States (Figure 7).

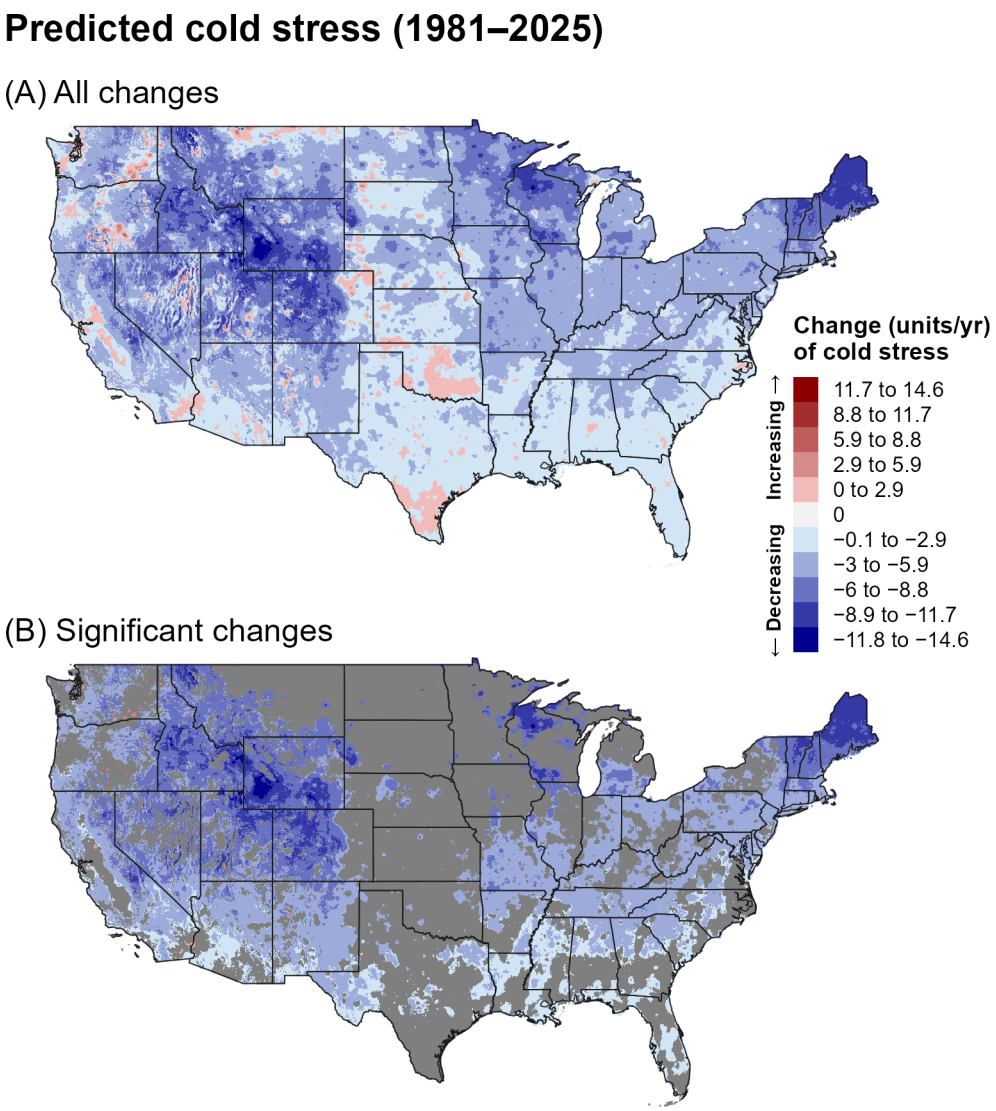


Figure 5. Changes (units/year) in predicted cold stress for STB across the CONUS over the entire 44-year period (1981–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant. Areas shown as 0 in (B) had significant trends but very small changes.

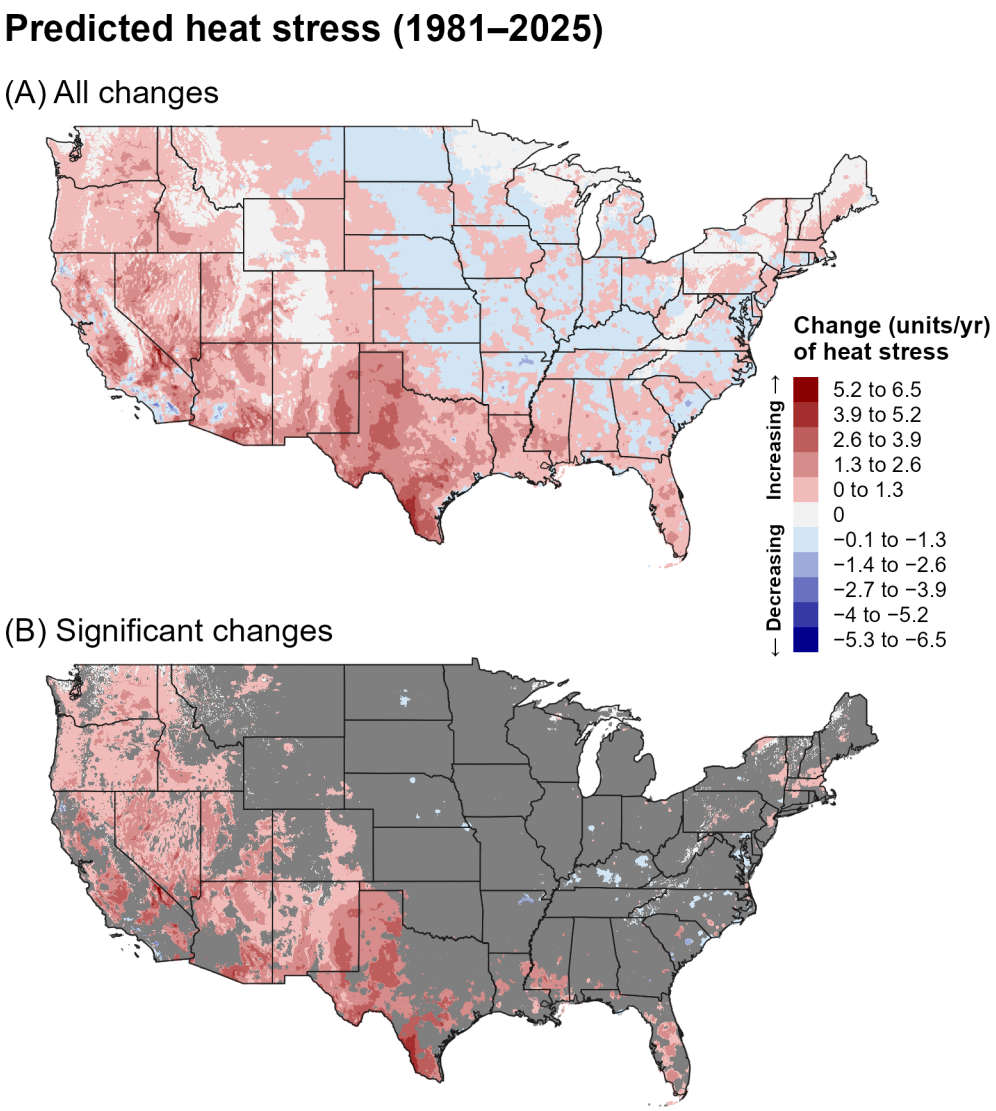


Figure 6. Changes (units/year) in predicted heat stress for STB across the CONUS over the entire 44-year period (1981–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant. Areas shown as 0 in (B) had significant trends but very small changes.

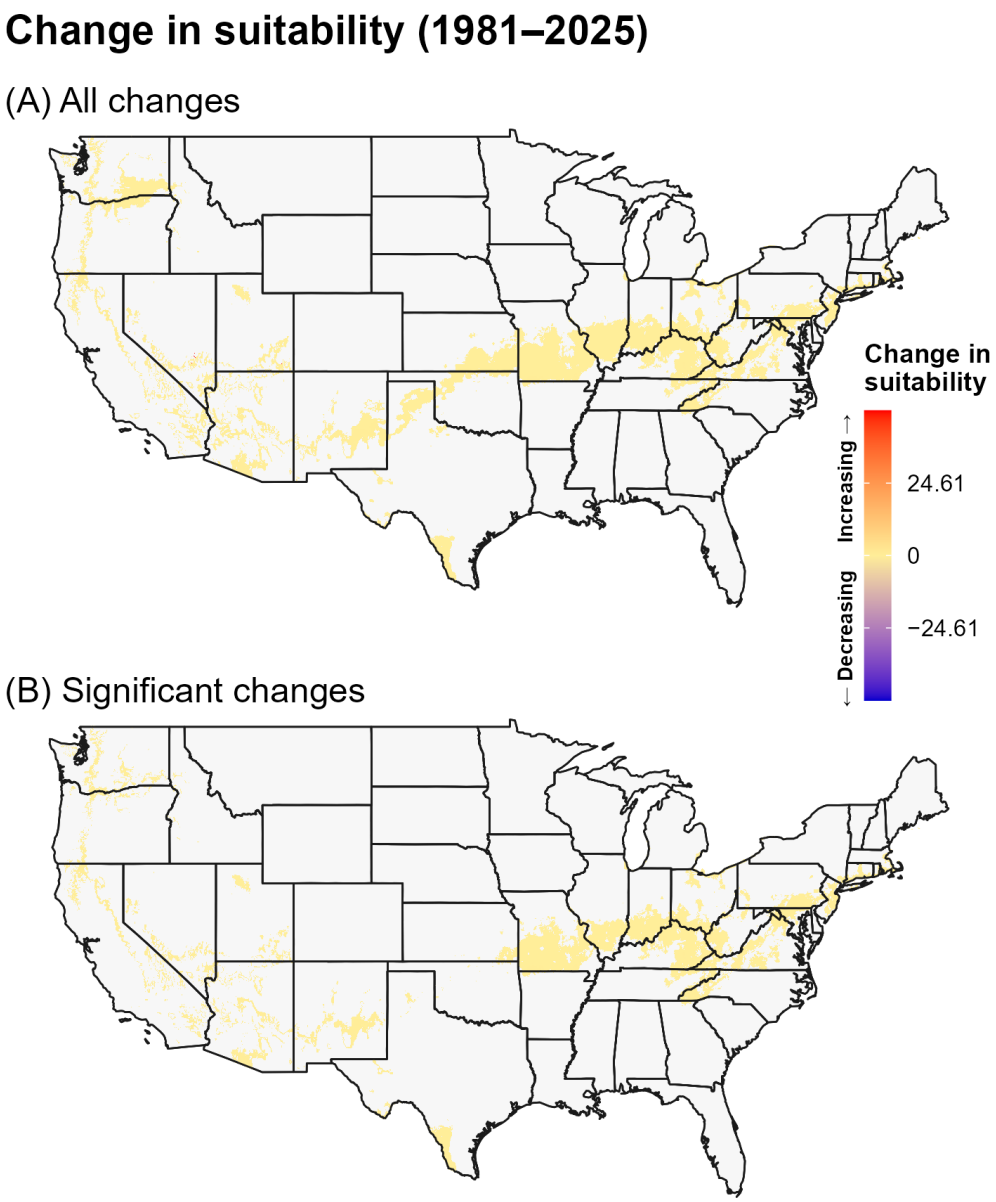


Figure 7. Changes in predicted climatic suitability for STB across the CONUS over the 44-year period (1981–2025). Suitability is defined as areas not excluded by climate stress. Positive changes (warmer colors) indicate an increased likelihood of establishment, whereas negative changes (cooler colors) indicate a decreased likelihood of establishment.

Recent period: 2001–2025

During 2001–2025, increases in predicted cold stress became prominent across portions of the northern and central interior, while heat stress continued to increase in the Southwest and southern Great Plains (Figure 8, Figure 9). Climatic-suitability increases remained concentrated in northern and central regions and were less extensive than over the full period (Figure 10).

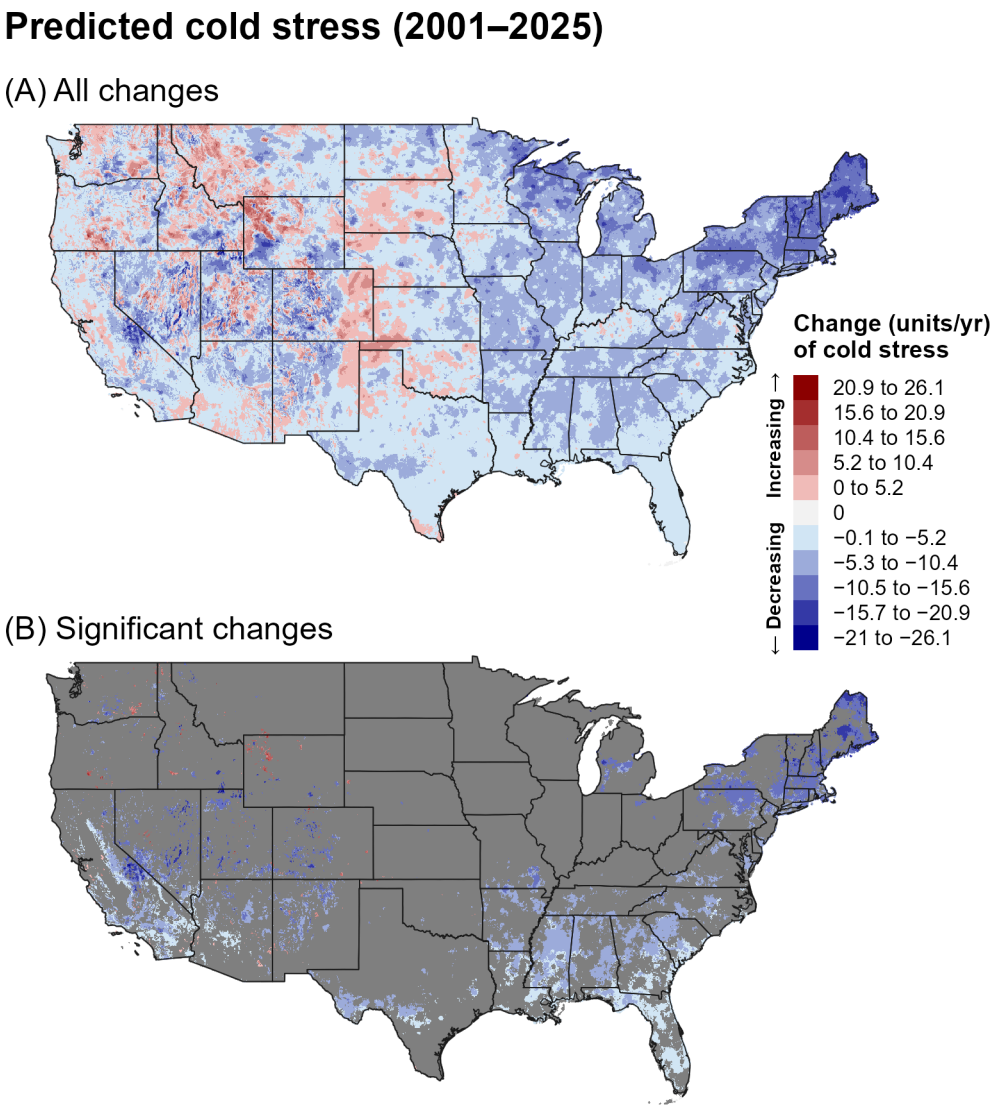


Figure 8. Changes (units/year) in predicted cold stress for STB across the CONUS over the recent 24-year period (2001–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant. Areas shown as 0 in (B) had significant trends but very small changes.

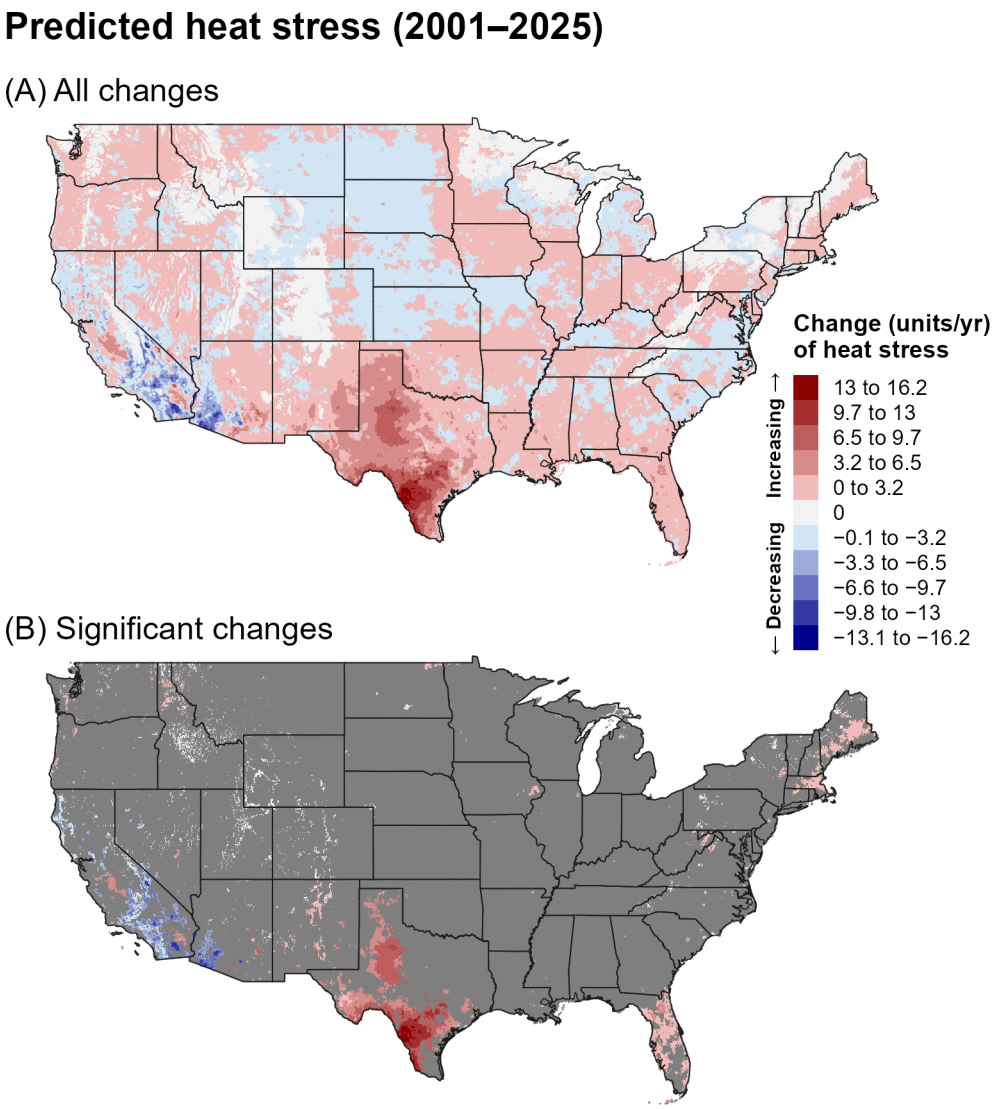


Figure 9. Changes (units/year) in predicted heat stress for STB across the CONUS over the recent 24-year period (2001–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant. Areas shown as 0 in (B) had significant trends but very small changes.

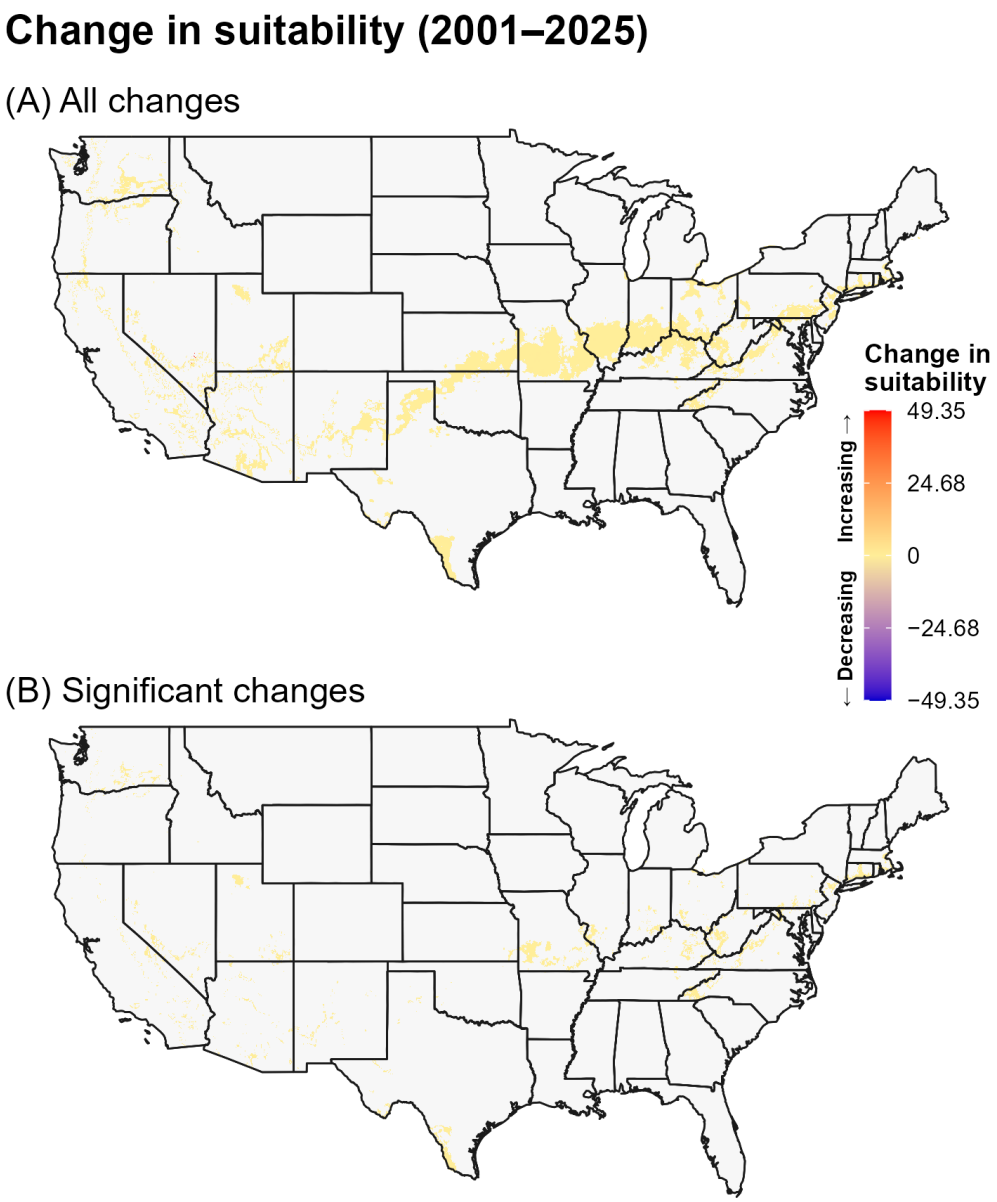


Figure 10. Changes in predicted climatic suitability for STB across the CONUS over the recent 24-year period (2001–2025). Suitability is defined as areas not excluded by climate stress. Positive changes (warmer colors) indicate an increased likelihood of establishment, whereas negative changes (cooler colors) indicate a decreased likelihood of establishment.

16C. Discussion

The STB results show a clear difference between the long-term and recent phenological patterns. Earlier emergence and egg hatch were common over 1981–2025, whereas later events became more prominent in the recent period across portions of the northern and central United States. This contrast suggests that recent regional climate variability has substantially influenced the short-term phenological signal and should be considered when interpreting current management timing.

The increase in predicted suitability across northern and central regions over the full period coincided with declining cold stress, suggesting that warming has reduced a temperature-related limitation in those areas. The

recent increase in cold stress and corresponding reduction in suitability changes indicate that this long-term expansion signal is less evident during 2001–2025.

The stronger and more spatially extensive trends in predicted phenology and climate stress over the full 44-year period may reflect both the larger sample size, which can reduce noise and improve detection of directional patterns, and differences in the underlying climate trends between the two time periods. In particular, the recent 24-year period may capture climate variability or regional trends that differ from those observed over the longer 44-year period. Thus, differences between the two periods should be interpreted as reflecting both long-term climate trends and shorter-term regional variability.

16D. References

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